

1.1

- a) Show that in the non relativistic limit the action S_{pp} has the usual nonrelativistic form, kinetic energy minus potential energy, with the potential energy being the rest mass.

$$S_{pp} = -m \int d\tau (-\dot{X}^\mu \dot{X}_\mu)^{1/2}, \quad \text{the metric of spacetime is } \eta_{\mu\nu} = \text{diag}(-1, 1, 1, \dots, 1)$$

$$\Rightarrow -\dot{X}^\mu \dot{X}_\mu = +\dot{X}^0 \dot{X}_0 - \dot{X}^a \dot{X}_a$$

Non relativistic limit means velocity $v^a = \dot{X}^a$ satisfies $v \ll 1$.

$\Rightarrow \dot{X}^a \dot{X}_a \ll \dot{X}^0 \dot{X}_0 \Rightarrow$ we can approximate using binomial approximation
 $(a+b)^\alpha = a^\alpha (1 + \frac{b}{a})^\alpha \approx a^\alpha (1 + \alpha \frac{b}{a}) = a^\alpha + \alpha a^{\alpha-1} b$

$$\Rightarrow (-\dot{X}^\mu \dot{X}_\mu)^{1/2} \approx (\dot{X}^0 \dot{X}_0)^{1/2} \left(1 - \frac{\dot{X}_a \dot{X}^a}{\dot{X}^0 \dot{X}_0}\right)^{1/2} \quad \parallel \dot{X}_a \dot{X}^a = v^2$$

$$\approx (\dot{X}^0 \dot{X}_0)^{1/2} \left(1 - \frac{1}{2} \frac{v^2}{\dot{X}^0 \dot{X}_0}\right)$$

$$\approx 1 - \frac{1}{2} v^2$$

$\parallel$ In the non-relativistic limit $\frac{\dot{X}^0}{\dot{X}_0} = 1$

$$\Rightarrow S_{pp} = -m \int d\tau (-\dot{X}^\mu \dot{X}_\mu)^{1/2} \approx \int d\tau \frac{1}{2} m v^2 - m = \int d\tau K - V,$$

where $K = \frac{1}{2} m v^2$ is the kinetic energy of the particle and $V = m$ is the potential energy.

- b) Show that for a string moving nonrelativistically, the Nambu-Goto action reduces to a kinetic term minus a potential term proportional to the length of the string. Show that the kinetic energy comes only from the transverse velocity of the string. Calculate mass per unit length, as determined from the potential term and also from the kinetic term.

Nambu Goto action:

$$S_{NG} = \int_M d\tau d\sigma \mathcal{L}_{NG}, \quad \text{where } \mathcal{L}_{NG} = -\frac{1}{2\pi\alpha'} (-\det h_{ab})^{1/2} \quad \parallel \begin{matrix} h_{ab} \\ = \partial_a X^\mu \partial_b X_\mu \end{matrix}$$

So,
$$S_{NG} = - \int_M dr d\sigma \frac{1}{2\pi\alpha'} (-\det \partial_a X^\mu \partial_b X_\mu)^{1/2}$$

Let's figure out $\det \partial_a X^\mu \partial_b X_\mu$ for non-relativistic case

$$\partial_a X^\mu \partial_b X_\mu = -\partial_a X^0 \partial_b X^0 + \partial_a X^i \partial_b X_i$$

Let's work in coordinates where $X^0 (= X^0(\tau))$

$$\det \partial_a X^\mu \partial_b X_\mu = \det (-\partial_a X^0 \partial_b X^0 + \partial_a X^i \partial_b X_i)$$

Note $X = X(\tau, \sigma)$
but also $X^0 = X(\tau)$

$\Rightarrow \partial_a X^0$ is
non-zero only for
 $a = \tau$

$$= \det \begin{bmatrix} -\dot{X}^0 \dot{X}^0 + \dot{X}^i \dot{X}_i & \dot{X}^i \partial_\sigma X_i \\ \partial_\sigma X^i \dot{X}_i & \partial_\sigma X^i \partial_\sigma X_i \end{bmatrix}$$

$$= (-\dot{X}^0 \dot{X}^0 + \dot{X}^i \dot{X}_i) \partial_\sigma X^j \partial_\sigma X_j - \partial_\sigma X^i \dot{X}_i \dot{X}^j \partial_\sigma X_j$$

$$= (-\dot{X}^0 \dot{X}^0 + \dot{X}^i \dot{X}_i) (\partial_\sigma X^j)^2 - (\partial_\sigma X^j \dot{X}_j)^2$$

$$\det \partial_a X^\mu \partial_b X_\mu = (-\dot{X}^0 \dot{X}^0 + \dot{X}^2) (X')^2 - (X' \cdot \dot{X})^2$$

Since $\dot{X}^i = \dot{X}_i$
 $\dot{X}^j \partial_\sigma X_j = \dot{X}_j \partial_\sigma X^j$

let's write $\partial_\sigma X^i = X'^i$

$$X' \cdot \dot{X} = X'^i \dot{X}_i$$

$$(X')^2 = X'^i X'_i$$

$$\dot{X}^2 = \dot{X}^i \dot{X}_i$$

Now let's apply the non-relativistic approximation:

$$\dot{X}^0 \approx 1, \quad \dot{X}^2 \approx v^2, \quad v^2 \ll 1$$

$$\Rightarrow -\det \partial_a X^\mu \partial_b X_\mu = (1 - v^2) (X')^2 + (X' \cdot \dot{X})^2$$

$$\Rightarrow \sqrt{-\det \text{hab}} = |X'| \sqrt{1 - v^2 + (X' \cdot \dot{X})^2 / (X')^2}$$

$$\begin{aligned} |X'| &= \sqrt{(X')^2} \\ \dot{X}^i &= v^i \\ \text{so } \frac{(X' \cdot \dot{X})^2}{(X')^2} &\ll 1 \end{aligned}$$

Using binomial approximation we get

$$\sqrt{-\det \text{hab}} \approx |X'| \left(1 - \frac{1}{2} (v^2 - \frac{(X' \cdot \dot{X})^2}{(X')^2}) \right)$$

$$\Rightarrow S_{NG} = - \frac{1}{2\pi\alpha'} \int_M dr d\sigma |X'| \left(1 - \frac{1}{2} v^2 + \frac{1}{2} \frac{(X' \cdot \dot{X})^2}{(X')^2} \right)$$

let's just
write $\dot{X} = v$

$$= \frac{1}{2\pi\alpha'} \int_M dr d\sigma \left(\frac{1}{2} |X'| v^2 - \frac{1}{2} \frac{(X' \cdot v)^2}{|X'|} - |X'| \right)$$

$$S_{NG} = \int_M d\tau d\sigma \left[\frac{1}{2} \frac{|X'|}{2\pi\alpha'} \left(v^2 - \frac{(X' \cdot v)^2}{|X'|^2} \right) - \frac{|X'|}{2\pi\alpha'} \right]$$

This is of the form $S_{NG} = \int_M d\tau d\sigma K - V$, where K is the kinetic term & V is the potential term.

Because kinetic term has $v^2 - \frac{(X' \cdot v)^2}{|X'|^2}$, the velocity that contributes is only the transverse velocity

$\Rightarrow$ IE $\frac{(X' \cdot v)^2}{|X'|^2}$ is the tangential velocity of string. Here by tangential, we mean tangential to the string itself

i.e. tangential to $\partial_\sigma X^i$ direction because $\partial_\sigma X^i v_i$ projects the total velocity to only its tangential component, which means

$v^2 - \frac{(X' \cdot v)^2}{|X'|^2}$ is transverse velocity. Note: $|X'| = \sqrt{X'^i X'^i}$

looking at the form of Kinetic & Potential terms and comparing to how it was in previous calculation involving point particle ($K = \frac{1}{2}mv^2$, $V = m$) we see that mass per unit unit of length is

$$m = \frac{|X'|}{2\pi\alpha'} \text{ which means}$$

$$S_{NG} = \int_M d\tau d\sigma \left[\frac{1}{2} m v_T^2 - m \right], \text{ where } v_T \text{ is transverse velocity.}$$

1.2 Show that classical equations of motion imply that ends of the open string move at speed of light.

Open strings: Neumann boundary condition: $\delta X^\mu(\tau, 0) = \delta X^\mu(\tau, \ell) = 0$

The classical EOM (equations of motion) are

$$\nabla^2 X^\mu = 0 \text{ and } T_{ab} = -\frac{1}{\alpha'} \left(\partial_a X^\mu \partial_b X_\mu - \frac{1}{2} \eta_{ab} \partial_c X^\mu \partial^c X_\mu \right) = 0$$

Let's choose a gauge where $\eta_{ab} = \eta_{ab} = \text{diag}(-1, 1)$ so that our calculation simplifies.

But $\eta_{ab} = \eta_{ab} = \text{diag}(-1, 1)$ but $\eta_{ab} = \eta_{ab} = \text{diag}(-1, 1)$ for a constant A .

From this gauge choice we get

$$T_{00} = -\frac{1}{\alpha'} (\dot{X}^\mu \dot{X}_\mu - \frac{1}{2} (-1) (-\dot{X}^\mu \dot{X}_\mu + X'^\mu X'_\mu))$$

$$= -\frac{1}{\alpha'} \frac{1}{2} (\dot{X}^\mu \dot{X}_\mu + X'^\mu X'_\mu) = 0$$

$$\Rightarrow \dot{X}^\mu \dot{X}_\mu + X'^\mu X'_\mu = 0$$

Similarly $T_{11} = -\frac{1}{\alpha'} (X'^\mu X'_\mu - \frac{1}{2} (-\dot{X}^\mu \dot{X}_\mu + X'^\mu X'_\mu)) = -\frac{1}{2\alpha'} (\dot{X}^\mu \dot{X}_\mu + X'^\mu X'_\mu)$

$$T_{01} = -\frac{1}{\alpha'} (\dot{X}^\mu X'_\mu) = 0$$

So we have the constraints

$$\dot{X}^\mu \dot{X}_\mu + X'^\mu X'_\mu = 0 \quad \text{and}$$

$$\dot{X}^\mu X'_\mu = 0$$

Now let's apply the boundary condition $X'_\mu = 0$ at $\sigma = 0, l$

$$\Rightarrow \dot{X}^\mu \dot{X}_\mu = 0 \quad \text{at} \quad \sigma = 0 \quad \& \quad \sigma = l \quad \text{i.e. ends of string}$$

$$\dot{X}^\mu \dot{X}_\mu \Big|_{\sigma=0, l} = 0 \Rightarrow \text{4-velocity } \dot{X}^\mu \text{ is null} \Rightarrow \text{the ends of the open string move at speed of light.}$$

$\Rightarrow$ Classical equations of motion imply that the ends of the open strings move at the speed of light.

1.3 For world-sheets with open boundary, show that

$$\chi = \frac{1}{4\pi} \int_M d\tau d\sigma (-\gamma)^{1/2} R + \frac{1}{2\pi} \int_{\partial M} ds \cdot k$$

is Weyl invariant. Here ds is the proper time along the boundary in the metric γ_{ab} , and k is the geodesic curvature of the boundary,

$$k = \pm t^a n_b \nabla_a t^b,$$

where t^a is a unit vector tangent to the boundary and n^a is an outward pointing unit vector orthogonal to t^a . The upper sign is for a Lorentzian world-sheet and the lower sign for a Euclidean world sheet.

Under Weyl transformation we have:

$$X'(\tau, \sigma) = X(\tau, \sigma) \quad \text{and}$$

$$\gamma'_{ab}(\tau, \sigma) = \exp(2w(\tau, \sigma)) \gamma_{ab}(\tau, \sigma)$$

for arbitrary $w(\tau, \sigma)$. Note that γ', X' here denotes transformed quantity and not a derivative.

Useful result from the chapter: $(-\gamma')^{1/2} R' = (-\gamma)^{1/2} (R - 2\nabla^2 w)$

Now let's calculate X' :

$$X' = \frac{1}{4\pi} \int_M d\tau d\sigma (-\gamma')^{1/2} R' + \frac{1}{2\pi} \int_{\partial M} ds' k'$$

Under

$$\Rightarrow \det h_{ab} = h_{\alpha\alpha} = h$$

$$ds = \sqrt{\pm h} d\lambda$$

where $h_{ab} = h_{\alpha\alpha}$ (1-dim metric) is the metric of the boundary and $d\lambda$ is the line element in the boundary. The upper sign is

for Lorentzian world-sheet & lower one for a Euclidean world sheet.

$\Rightarrow$

under Weyl rescaling $h \rightarrow e^{2w(\tau, \sigma)} h = h'$

$$\Rightarrow ds' = e^w ds$$

As for $k = \pm t^a n_b \nabla_a t^b$ we get $k \rightarrow k' = \pm t'^a n'_b \nabla'_a t'^b$, where $t'^a = e^{-w} t^a$, $t'_a = e^w t_a$ and same for n_b , because t'^a should also be a unit tangent vector when metric is γ'_{ab} ie

$$t'^a t'_a = t^a t_a \Rightarrow \gamma'_{ab} t'^a t'^b = \gamma_{ab} t^a t^b$$

$$\Rightarrow k' = \pm e^{-w} t^a e^w n_b \nabla'_a t'^b$$

$$\nabla_a t^b = \partial_a t^b + \Gamma_{ac}^b t^c$$

$$\Rightarrow \nabla'_a t'^b = \partial_a t'^b + \Gamma'_{ac}^b t'^c$$

$$\Gamma'_{ac}^b = \frac{1}{2} \gamma'^{bd} (\partial_a \gamma'_{cd} + \partial_c \gamma'_{da} - \partial_d \gamma'_{ac})$$

$$\nabla'_a t'^b = \partial_a (e^{-w} t^b) + \frac{1}{2} \gamma'^{bd} (\partial_a \gamma'_{cd} + \partial_c \gamma'_{da} - \partial_d \gamma'_{ac}) t'^c$$

$$= -(\partial_a w) e^{-w} t^b + e^{-w} \partial_a t^b + \frac{1}{2} \gamma'^{bd} (2(\gamma_{cd} \partial_a w + \gamma_{da} \partial_c w - \gamma_{ac} \partial_d w) + \partial_a \gamma_{cd} + \partial_c \gamma_{da} - \partial_d \gamma_{ac}) e^{-w} t^c$$

$$\gamma'^{bd} = e^{-2w} \gamma^{bd}$$

$$\partial_a (\gamma'_{cd}) = \partial_a (e^{2w} \gamma_{cd}) = (2\partial_a w \gamma_{cd} + \partial_a \gamma_{cd}) e^{2w}$$

$$= e^{-w} (\partial_a t^b + \Gamma_{ac}^b t^c) - e^{-w} t^b \partial_a w + \gamma^{bd} (\gamma_{cd} \partial_a w + \gamma_{da} \partial_c w - \gamma_{ac} \partial_d w) e^{-w} t^c$$

Now let's contract by substituting this into k'

$$k' = \pm t'^a n_b [e^{-w} \nabla_a t^b - e^{-w} t^b \partial_a w + \gamma^{bd} (\gamma_{cd} \partial_a w + \gamma_{da} \partial_c w - \gamma_{ac} \partial_d w) e^{-w} t^c]$$

$$k' = e^{-w} k \pm [e^{-w} t^a n_b t^b \partial_a w + t^a n^d (t_d \partial_a w + \gamma_{da} t^c \partial_c w - \gamma_{ac} t^c \partial_d w) e^{-w}]$$

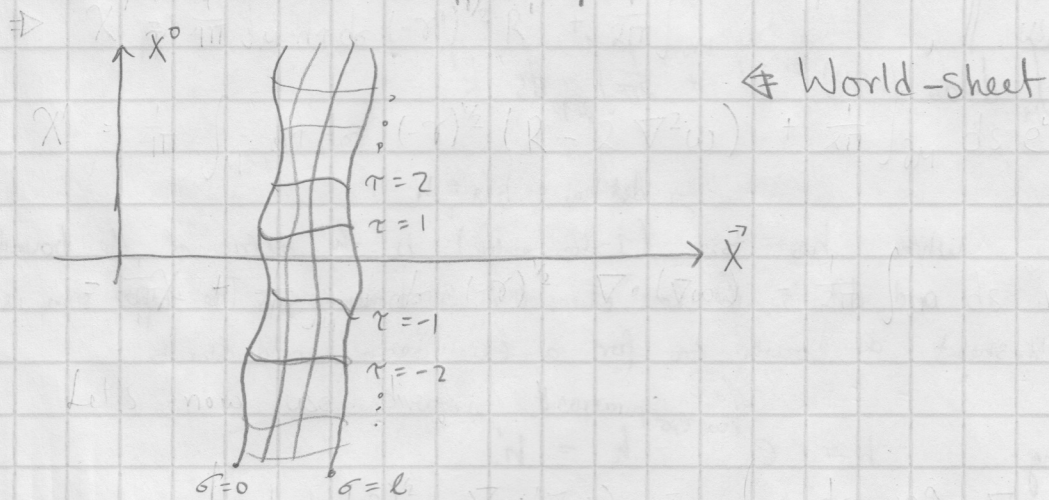
We can drop all term where n and t are contracted since $n^a t_a = 0$

$$\Rightarrow k' = e^{-\omega} k \pm e^{-\omega} (-t^a t_a) n^a \partial_a \omega$$

For Euclidean world-sheet $t^a t_a = 1 \Rightarrow k' = e^{-\omega} k + e^{-\omega} n^a \partial_a \omega$

For Lorentz world-sheet $t^a t_a = \pm 1$ for spatial or timelike boundary,

Our construction is the following:



We have $-\infty < \tau < \infty$ and $0 \leq \sigma \leq l$ so the boundary of string world sheet on either side ($\sigma=0$ or $\sigma=l$) is time-like so the tangent vectors t^a are timelike $\Rightarrow t^a t_a = -1$

So for Lorentz world-sheet $t^a t_a = -1 \Rightarrow k = e^{-\omega} k + e^{-\omega} n^a \partial_a \omega$

So in both cases (Euclid vs Lorentz) $k = e^{-\omega} k + e^{-\omega} n^a \partial_a \omega$

Let's substitute this to χ' :

$$\chi' = \frac{1}{4\pi} \int_M d\tau d\sigma (-\gamma)^{1/2} R' + \frac{1}{2\pi} \int_{\partial M} ds' k'$$

$$\chi' = \chi - \frac{1}{2\pi} \int_M dV \nabla \cdot (\nabla \omega) + \frac{1}{2\pi} \int_{\partial M} ds n^a \partial_a \omega$$

Let's now use divergence theorem $\int_V dV \nabla \cdot F = \int_{\partial V} ds n^a F_a$

$$\Rightarrow \chi' = \chi - \frac{1}{2\pi} \int_{\partial M} ds n^a \partial_a \omega + \frac{1}{2\pi} \int_{\partial M} ds n^a \partial_a \omega$$

$\chi' = \chi \Rightarrow$ For a world sheet boundary

$$\chi = \frac{1}{4\pi} \int_M d\tau d\sigma (-\gamma)^{1/2} R + \frac{1}{2\pi} \int_{\partial M} ds k \text{ is Weyl-invariant.}$$

$$\left\{ \begin{array}{l} k' = e^{-\omega} (k + n^a \partial_a \omega) \\ ds' = e^{\omega} ds \\ (-\gamma')^{1/2} R' = (-\gamma)^{1/2} (R - 2 \nabla^2 \omega) \\ dV = d\tau d\sigma (-\gamma)^{1/2} \text{ is the volume element} \end{array} \right.$$

$\left\{ \begin{array}{l} \omega \text{ is a scalar} \\ \Rightarrow \nabla_a \omega = \partial_a \omega \\ \text{the integrals cancel out} \end{array} \right.$